

End

dinatural transformation

Let $F, G: C^{\text{op}} \times C \rightarrow D$ be functors. A **dinatural transformation** from F to G , sometimes written

$$\alpha: F \overset{\bullet}{\rightarrow} G,$$

consists of a collection of morphisms

$$\alpha_c: F(c, c) \rightarrow G(c, c)$$

such that for every morphism $f: c \rightarrow c'$ in C ,

$$G(c, f)\alpha_c F(f, c) = G(f, c')\alpha_{c'} F(c', f): F(c', c) \rightarrow G(c, c')$$

If drawn out as a commutative diagram, this becomes a “hexagon identity”:

$$\begin{array}{ccccc} & F(c, c) & \xrightarrow{\alpha_c} & G(c, c) & \\ & \nearrow F(f, c) & & \searrow G(c, f) & \\ F(c', c) & & & & G(c, c') \\ & \searrow F(c', f) & & \nearrow G(f, c') & \\ & F(c', c') & \xrightarrow{\alpha_{c'}} & G(c', c') & \end{array}$$

Special cases

If F and G both factor through the projection $C^{\text{op}} \times C \rightarrow C$, then the notion reduces to an ordinary natural transformation, and similarly if they both factor through C^{op} .

If F factors through C while G factors through C^{op} , then we obtain a notion of natural transformation from a covariant functor to a contravariant one, and dually.

If G is constant, the hexagon identity reduces to the “domain” version of extranaturality, involving a commutative square of the form

$$\alpha_c F(f, c) = \alpha_{c'} F(c', f) : F(c', c) \rightarrow G$$

where G is constant with respect to the argument c . Similarly, if F is constant, it yields the “codomain” version with a commutative square of the form

$$G(c, f) \alpha_c = G(f, c') \alpha_{c'} : F \rightarrow G(c, c')$$

when F is constant with respect to the argument c .

Wedge

设：

$$H : \mathcal{C}^{op} \times \mathcal{C} \rightarrow \mathcal{E}.$$

以及 constant bifunctor：

$$\Delta X : \mathcal{C}^{op} \times \mathcal{C} \rightarrow \mathcal{E}$$

定义：

$$\Delta X(c, c') = X.$$

那么：

$$\boxed{\omega : \Delta X \Rightarrow H}$$

就是一个 dinatural transformation。

它的 component 是：

$$\omega_c : X \rightarrow H(c, c).$$

dinaturality condition 变成：

$$\boxed{H(f, 1_{c'}) \circ \omega_{c'} = H(1_c, f) \circ \omega_c}$$

这正是 wedge 的定义。

所以：

$$\boxed{\text{wedge} = \text{dinatural transformation } \Delta X \Rightarrow H}$$

Cowedge

如果：

$$\iota : H \Rightarrow \Delta X,$$

那么 component：

$$\iota_c : H(c, c) \rightarrow X.$$

dinaturality condition 变成：

$$\iota_c \circ H(1_c, f) = \iota_{c'} \circ H(f, 1_{c'})$$

这就是 **cowedge**。

所以：

$$\text{cowedge} = \text{dinatural transformation } H \Rightarrow \Delta X$$

end / coend

给定:

$$H : \mathcal{C}^{op} \times \mathcal{C} \rightarrow \mathcal{E}.$$

End

一个 end 是一个 universal dinatural transformation:

$$\boxed{\omega : \Delta \left(\int_c H(c, c) \right) \Rightarrow H}$$

即:

$$\omega_c : \int_c H(c, c) \rightarrow H(c, c).$$

它 universal 的意思是:

对于任意:

$$\alpha : \Delta X \Rightarrow H,$$

存在唯一:

$$u : X \rightarrow \int_c H(c, c)$$

使:

$$\boxed{\alpha = \omega \circ \Delta u}$$

Coend

一个 coend 是一个 universal dinatural transformation :

$$\boxed{\iota : H \rightrightarrows \Delta \left(\int^c H(c, c) \right)}$$

即:

$$\iota_c : H(c, c) \rightarrow \int^c H(c, c)$$

对于任意:

$$\alpha : H \rightrightarrows \Delta X,$$

存在唯一:

$$u : \int^c H(c, c) \rightarrow X$$

使:

$$\boxed{\alpha = \Delta u \circ \iota}$$

典型例子

- end 给出自然变换
- Yoneda 引理的 end 形式
- Yoneda 引理的 coend 形式
- Kan extension
- Geometric realization

优点

- End/coend 最大的优势：把“自然性条件”打包掉。
- End/coend 将 universal construction 转化成一种可进行代数操作的表达式。

End/coend 是范畴论的“积分/求和演算”。

而 Kan extension 更像是你要解决的 universal problem:

Kan extension: 我要构造什么?

End/coend: 这个 universal object 如何把它写成一个可操作的公式?

参考文献 I



`https://ncatlab.org/nlab/show/dinatural+transformation`



`https://chatgpt.com/s/t_6a96dbb328ec81919d35598bf6fee18d`